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ADVANCED WEB DESIGN DEVELOPMENT AND CONTENT MANAGEMENT (IMS566)

GROUP ASSIGNMENT :

UiTM LETTER MANAGEMENT SYSTEM (e-LETTER PORTAL)

CLASS : SI2624A

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TABLE OF CONTENT

1.0 Introduction....1
2.0 GitHub Repository Link....1
3.0 Entity-Relationship Diagram (ERD)....2
4.0 System Requirements....3
5.0 User Interface Overview....7
5.1 User Interface Layout.... 8
5.2 Student Dashboard....9
5.3 Letter Request Form Interface....10
5.4 Administrator Dashboard.... 11
5.5 Navigation Structure.... 12
6.0 Features & Functionalities....14
6.1 User Authentication and Role Management.... 14
6.2 Student Letter Request Submission.... 16
6.3 Letter Request Management.... 17
6.4 Letter Template Management.... 18
6.5 Automated Letter Generation.... 19
6.6 Application Tracking....20
7.0 Workflow of Form.... 21
7.1 View Letter Requests....21
7.2 Update Letter Request....22
7.3 Delete Letter Request....23
7.4 View Letter Templates.... 24
7.5 Create a New Letter Template....24
7.6 Update Letter Template.... 26
7.7 Delete Letter Template....27
8.0 Team Roles & Contributions....28
9.0 Contact Information....29
10.0 Conclusion & Reflection.... 30

1.0 Introduction

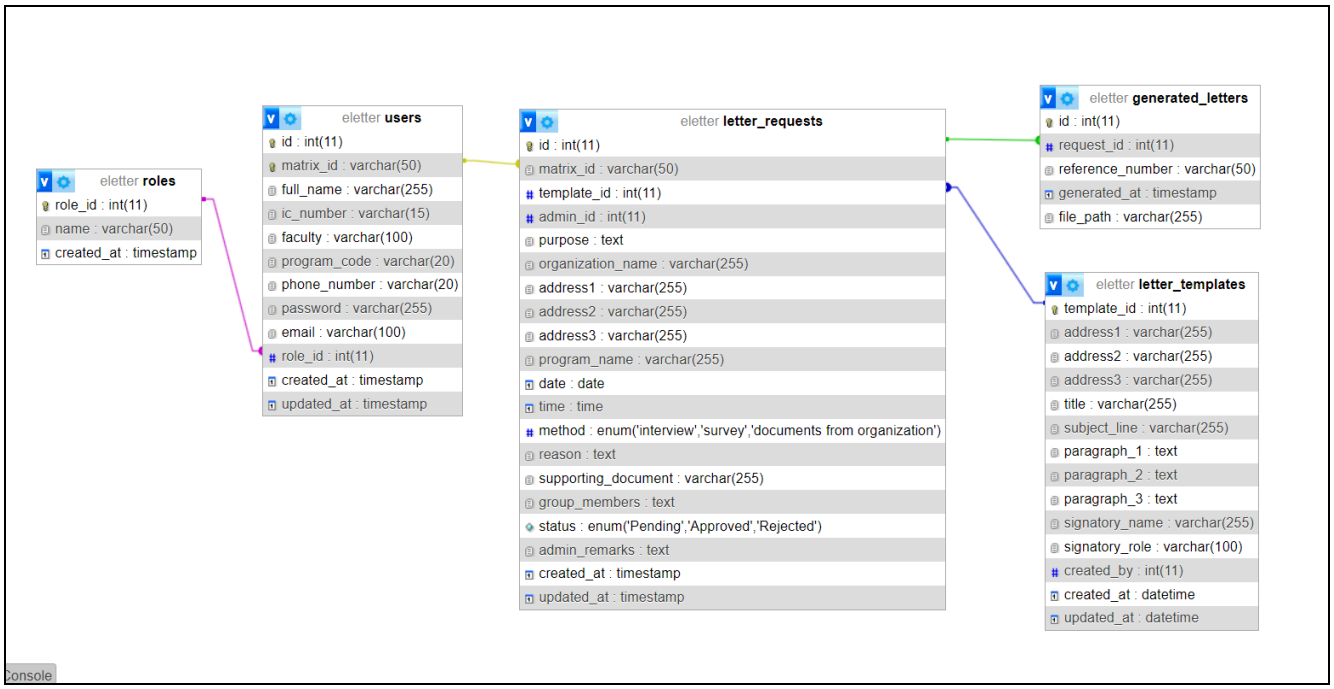
Our project is a web-based Letter Management System for Hal Ehwal Pelajar (HEP) UiTM. Students only need to fill in the required information using an online form instead of submitting the draft letters via Microsoft Word and Google Forms prepared by the HEP itself. The system will automatically generate letters and send them to HEP for review and approval. Staff can edit the content if necessary before issuing an official letter.

The purpose of this system is to simplify and digitize the letter application process for students and HEP. It provides a centralized digital record of all applications, enabling UiTM to analyze student activities, event participation, track frequently visited companies and support future collaborations with industry partners, while reducing administrative workload and increasing processing efficiency.

2.0 GitHub Repository Link

The full source code for this project, including documentation and version history, can be accessed at the repository link below:
https://github.com/Zepalien/eLetterUiTM

3.0 Entity-Relationship Diagram (ERD)



3.1 Entities

I. Roles : Students or Admin

II. Users : Store user personal information

III. Letter Templates : Store official letter templates used by HEP

IV. Letter Request : Store letter application that has been submitted by the students
Allow user to track approval letter

V. Generated Letters : Store the final approved letter

3.2 Relationship

I. **1:M - One to Many**

- One Roles to Many Users
- One Users to Many Letter Requests
- One Letter Templates to Many Letter Requests

II. **1:1 - One to One**

- One Letter Request to One Generated Letter

4.0 System Requirements

The UiTM Letter Management System (e-LETTER Portal) is the first web-based system to automate the application of letters among students and the Hal Ehwal Pelajar (HEP). A set of supported platform, environment and other supporting dependencies software are essential to be installed and provided before the UiTM Letter Management System is deployed.

Software Component	**Description**
Operating System	The system is compatible with Windows 10/11 and other operating systems that support web development environments.
Web Browser	The system can be accessed using modern web browsers such as Google Chrome, Microsoft Edge, and more to ensure proper interface rendering and functionality.
Development Environment	Visual Studio Code is used as the primary code editor for developing and managing the system files.
Front-End Technologies	HTML, CSS and JavaScript is used to develop the user interfaces of the systems and their interactive part.
Back-End Technologies	PHP has been utilized for manipulating operations at the server-side, as well as processing user inputs.
Database Management System	MYSQL is used to store and manage system data including user information, letter request, letter templates, and generated letters.
Local Server Environment	XAMPP is used as a local development server to run the system and establish database connectivity.
Version Control System	GitHub is used to manage source code, track project updates, and support collaboration among team members.

4.2 Version Compatibility

The software versions used throughout the system must be compatible so that the various system components will work well together and avoid technical conflicts. Software versions which are compatible with one another help to make the system run stable and also avoid errors from occurring at run-time so that the user is able to get in and work in the system.

Component	Recommended Version	Purpose
Operating system	Windows 10 / Windows 11	Provides the required environment for system development and execution.
PHP Environment	PHP 8.0 and above	Supports server-side processing and system logic execution.
MySQL Database	MySQL 5.7 and above	Provides reliable database storage and management capabilities.
XAMPP	Latest stable version	Provides Apache server and MySQL environment for local testing.
Web Browser	Latest version of Google Chrome, Microsoft Edge	Ensures proper display and interaction with the system interface.
Visual Studio Code	Latest stable version	Supports source code development and project management.

The chosen versions ensure that your system is able to handle your database needs and your web applications as well as deliver an seamless and integrated user experience on any of your users devices.

4.3 System Dependencies

In order to have all the functions in the UiTM Letter Management System work properly, the following components are needed to install as dependencies in the UiTM Letter Management System. This system depends on these external components for the purpose of functioning properly.

1. Database Connectivity Dependency

The system needs a continuous connection between the web application and the MySQL database. All the important information is stored within the database; the most important information are the user records, application letter submissions, application letter templates and generated documents. The users cannot submit application records nor can they retrieve previous requests, monitor previous applications or submit new application records without a connection to the database.

2. Web Server Dependency

The system relies on the web server to send and process user requests as well as to execute all system functions. A web server within the local environment provides web page information and function executions as a part of the application testing stage before the system is deployed.

3. Authentication and User Role Dependency

The system needs an authentication mechanism to recognize users and grant them access to the system, and also as the system needs students and HEP staff to use the system, the users will require different privileges. For example, students are restricted to submit application letters, monitor their submitted applications and see their application's progress. On the other hand, HEP staff or administrators are restricted to see student's applications and approve or reject their applications.

4. Browser Compatibility Dependency

The system relies on browser compatibility for students and HEP staff to be able to access the platform without difficulty in terms of presentation or function. This covers every interactive part of the web page, including buttons, navigation menus, input boxes, display of form data and the generation of documents.

5. Internet and Network Dependency

The internet connection is crucial because the system is web based. This would allow students and HEP staff members to communicate with the system efficiently and conduct their business, in terms of submitting requests and administrating tasks on the system.

5.0 User Interface Overview

The user interface (UI) of UiTM Letter Management System (e-Letter Portal) is aimed at facilitating a straightforward, orderly and user-friendly environment to be operated by the students as well as administrators. Given that the system has different types of users with the differences of their responsibilities, the user interface is organised according to their roles so that relevant functionalities can be accessed by the appropriate user in an efficient manner. The UI design's goal is to make the process of application for letters to be less complicated by reducing unnecessary process steps, enhancing access to information and making it more obvious with clearer process flow to be performed by users. For the students, interfaces are provided to enable them to make letter applications and trace back their application, whereas for administrators, a management interface is provided for the review and process of the application and to allow the approval for the letter application.

The interface design focuses on three main principles, which are:

- **Usability:** Our interface allows users to find the required menus and access the functions in a very clear way. Navigation in the menus and forms should not lead to difficulties for students and teachers. Functionality should not cause a difficulty for the user's tasks.

- **Consistency:** All of our menus and information should have the same appearance and behavior to give our users an easy use experience. This affects appearance structure, flow of page, and more.

- **Efficiency:** Automation of some work can reduce the manual process by having direct submission of the applications online and the requests can be handled through the system in a very organized way by the administrative staff.

5.1 User Interface Layout

UiTM letter management system consists of main components of interface which will assist the overall workflow of the UiTM letter management system. These interfaces have been designed by considering the needs of users which are students and administrators.

5.1.1 Login Interface

The login screen is the primary entry point for users of the system. Users will enter their authentication details in this section before they can enter the system. Based on the role assigned to a particular user account, the system will authenticate the users and will guide them to their respective views (student dashboard or management dashboard). If the user is a student then they will be provided access to the student's dashboard from where they can apply and check the status of their letters. If the user is an admin then they will be redirected to the management dashboard to review and verify applications. The login screen is a crucial element in a student letter application system as it safeguards and restricts the access to the personal student's data and the system in general.

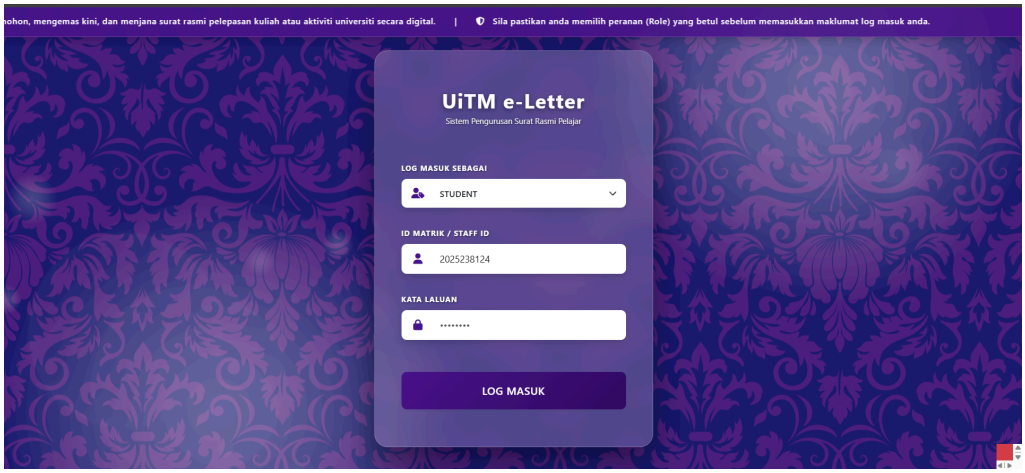


Figure 2: Login Interface

The interface supports the following functions:

- **User authentication:**
  Verifies user credentials to ensure that only registered users can access the system.
- **Role Identification:**
  Identifies whether the user is a student or administrator and provides access to relevant system features based on their role.
- **Secure access control:**
  Protects user information and prevents unauthorized access to the system.

5.2 Student Dashboard

Student Dashboard has all the main features that students need for handling their letter application and request. The student will use this dashboard to make a new request, check for past letters submitted and see the status of each application that was made for their letters. This dashboard gives a good and complete view of students' use of the system, as no multiple pages would be needed for application submission and verification; instead, students can make all of their letter request actions from just one page.

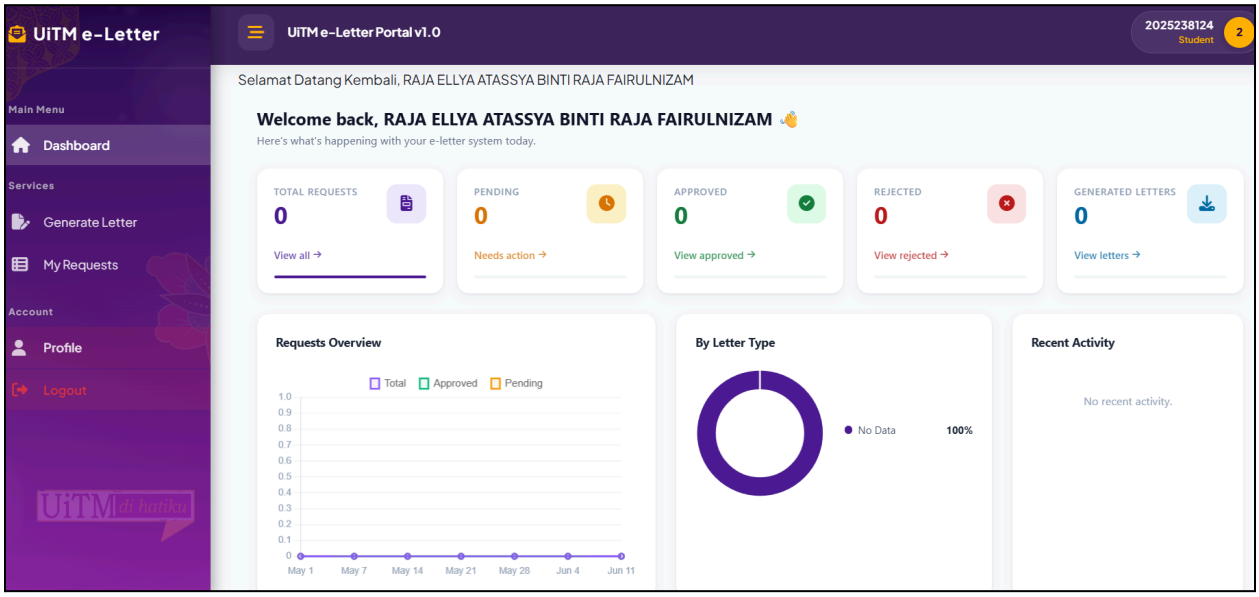


Figure 3: Student Dashboard

The interface supports the following functions:

- **Letter Request Access:**
  Provides access to student's application form for letter request and allows the submission of new requests by students via the system.

- **Application Monitoring:**
  The application monitoring feature allows students to view the progress of their applied applications. They can also view the current status of submitted applications in this tab.

- **Generated letter Access**:
  Students are allowed access to approved letters that were created within the system.

5.3 Letter Request Form Interface

The student can utilize the Letter request form interface to give their particulars as the mandatory information that is required for the processing of the official letters. The system had provided the form to replace students who draft the letter documentation via external program before. The form structure is guaranteed that the information is compiled consistently and it also ensures all the fields are filled up by the students before the form is submitted to HEP.

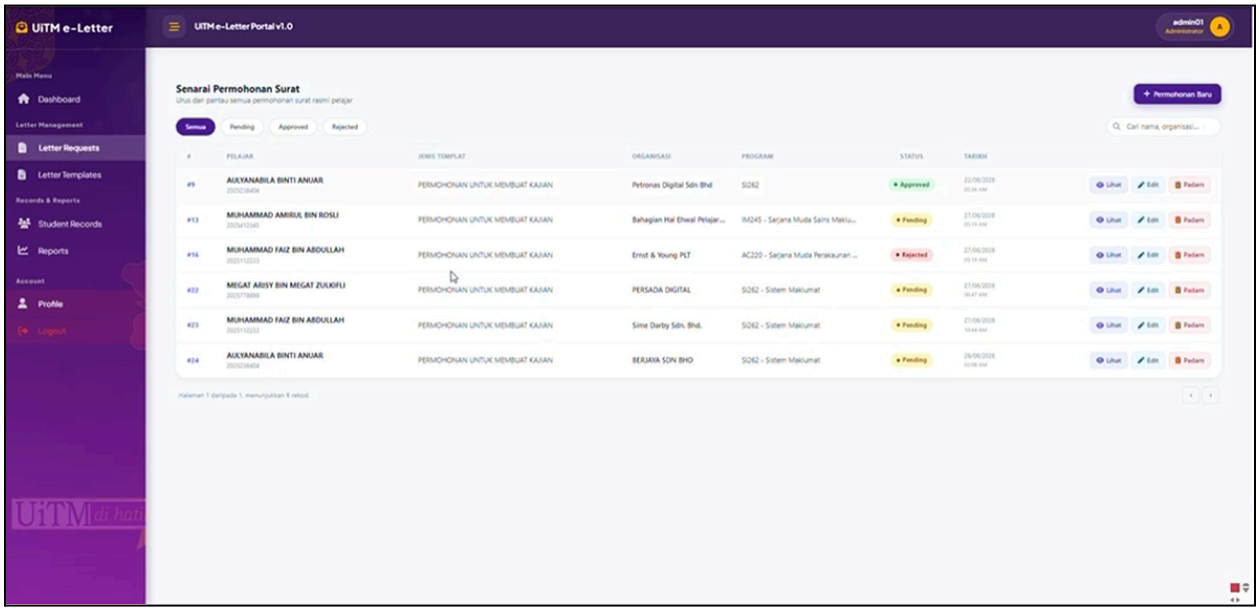


Figure 4: Letter Request Form Interface

The letter request form interface supports the following functions:

- **Information Submission:**
  students fill in information such as students information, activities information, reason of writing a letter and other information.

- **Data validation:**
  Make sure that fields that need information have been completed before the request is sent.

- **Application Storage:**
  The request details are input to the database within the store by them.

5.4 Administrator Dashboard

The Administrator Dashboard HEP users should access to manage the letter requests submitted by students. Administrators will use this tool to review and manage the letter requests that have been submitted by students, and in order to approve, deny, or update letter requests. All requests can be accessed via a single centralized view of all the letter request workflow for administrative purposes.

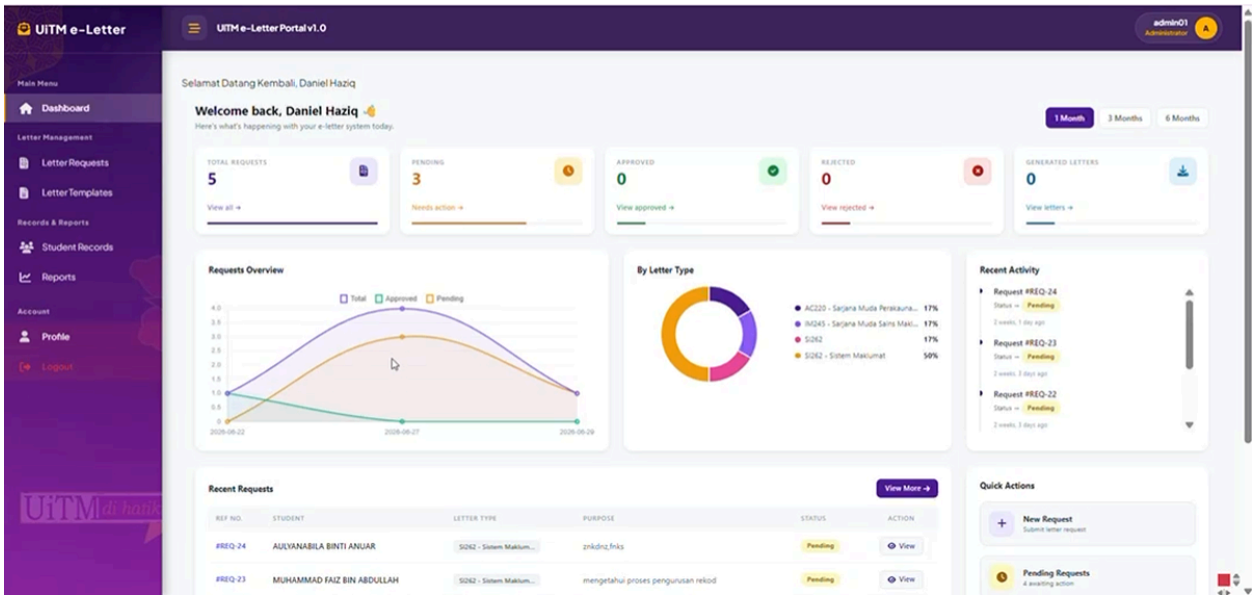


Figure 5: Administrator Dashboard

The administrator dashboard supports the following functions:

- **Application Review:**
  Students can view and check their outstanding submitted letters requests for Administrators to see and review student's submitted letters requests.
- **Request Management:**
  Allows administrators to update, edit or maintain application details as needed.

- **Approval Processing:**
  Enables Administrators to approve the request prior to producing the official letters.

5.5 Navigation Structure

5.5.1 Student Navigation Flow

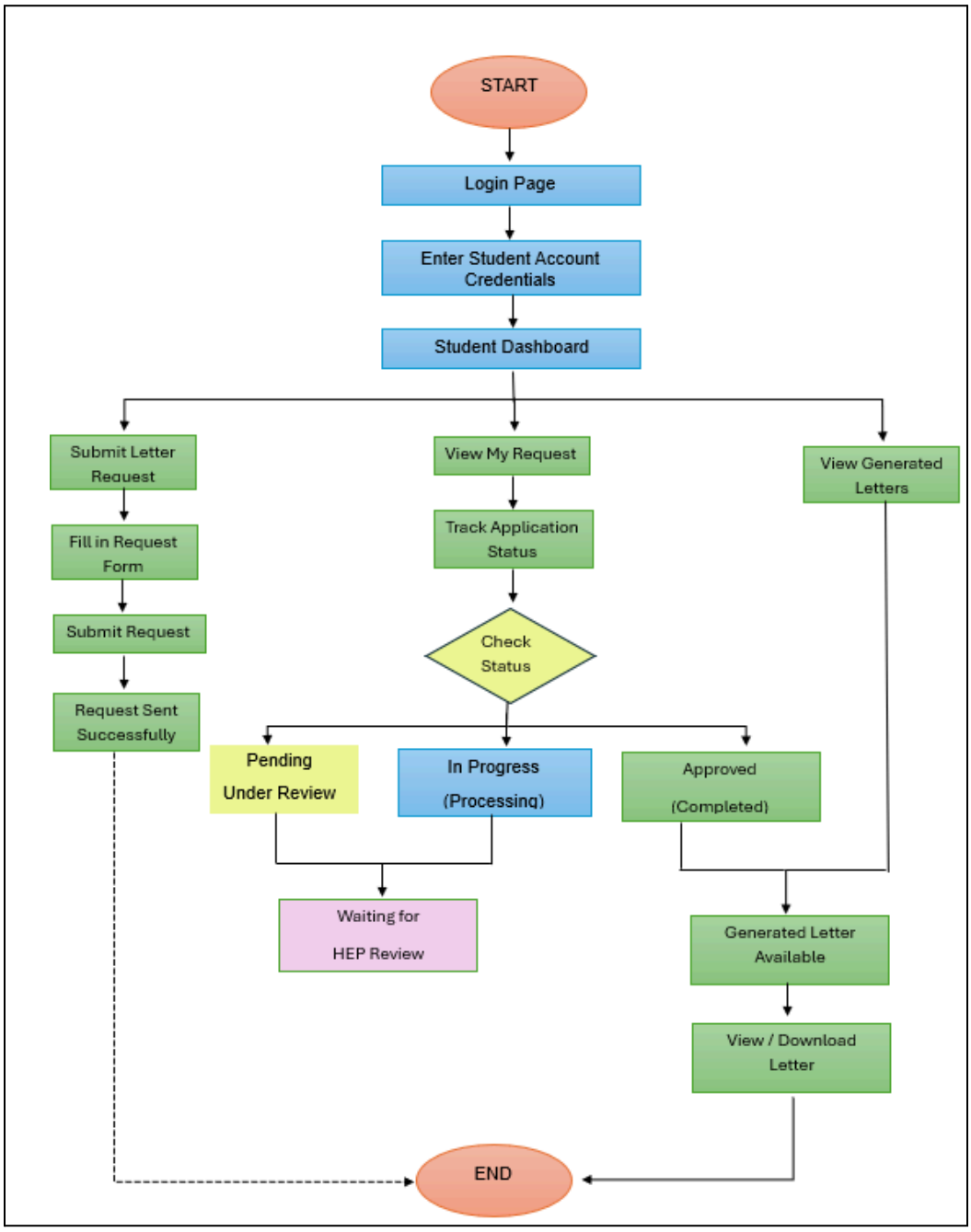


Figure 6: Student Navigation Flowchart

The navigation path for students starts with logging into the system through their respective login credential. Post the successful validation of the students log in, the user gets directed towards the student portal which displays various functions available to the students. A student

can fill out the application form to send out a new request for a letter and once sent, the same will be saved on the system and reviewed by HEP members. Students are kept notified about the application status and the resultant letter generated, upon acceptance of the application.

5.5.2 Administrator Navigation Flow

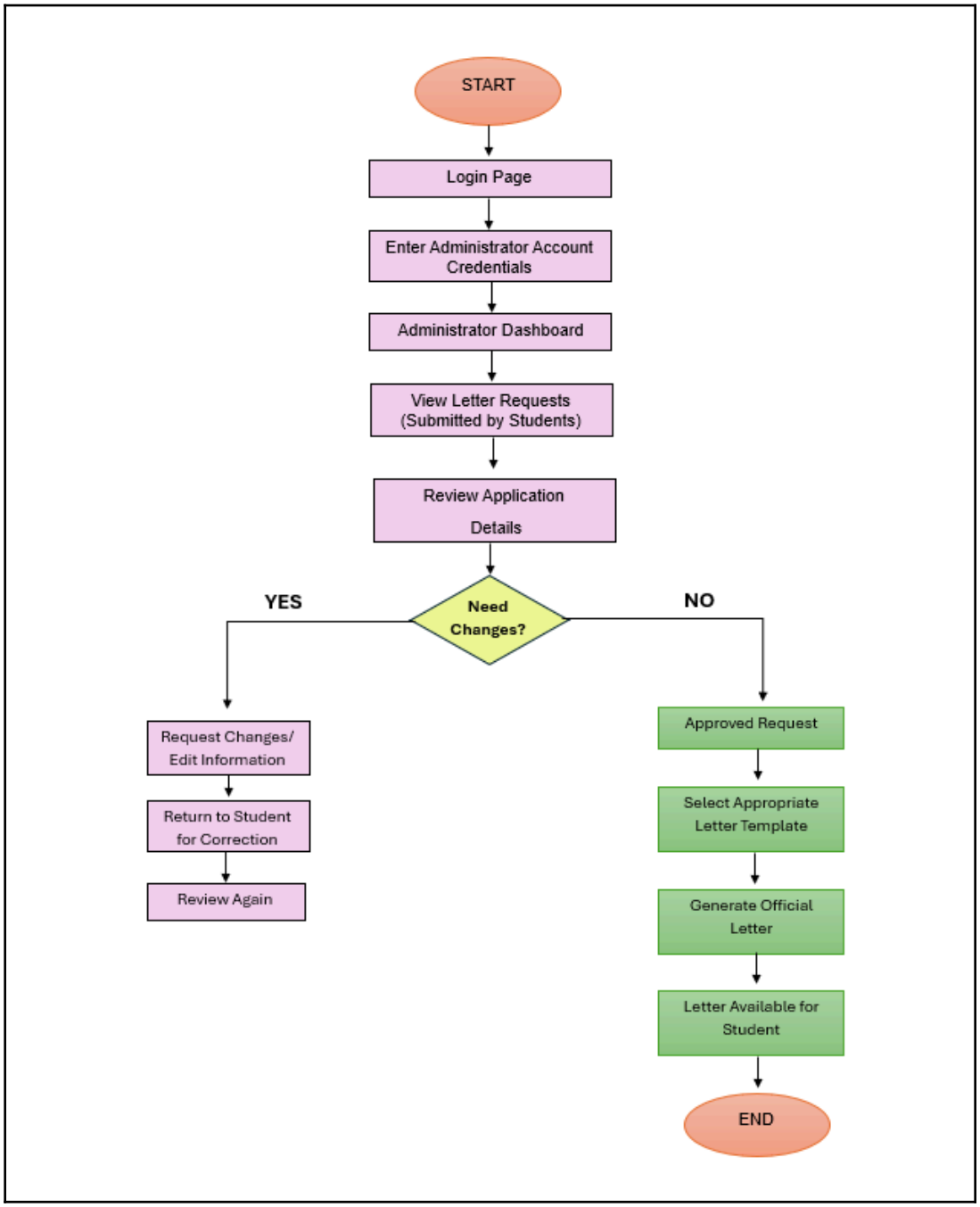


Figure 7: Administrator Navigation Flowchart

Administrator navigation flow starts from when the HEP Staff logs in to the system via Administrator Login and will land on a dashboard view from where all submitted student letter requests and student application data can be seen and checked. If modification is required in the application, the administrator can edit the application information before approval, and if everything is in place, the system will then automatically generate an official letter that the student can download.

6.0 Features & Functionalities

UiTM Letter Management System (e-Letter Portal) has few functions and features to manage the applications of student letters online, which are aimed to facilitate the application and process of official letters between students and the Hal Ehwal Pelajar (HEP). Previously, the student was supposed to make a drafted letter and submit it in a separate platform and method. This system would benefit students to do application in an electronic method and the administrator manage it in a consolidated platform. The main functionalities of the system focus on improving efficiency, reducing manual documentation, maintaining organized records, and providing easier access to letter application information.

6.1 User Authentication and Role Management

This functionality provides security for the users when they can access the system based on the assigned roles. As the system has 2 types of users, which are the student, the administration, different levels of access should be given to enable users to carry out the job related to their duties only. The students can use this to submit the letter application and check on the status of the application whereas the administration is to approve, check and modify the submitted letter application.

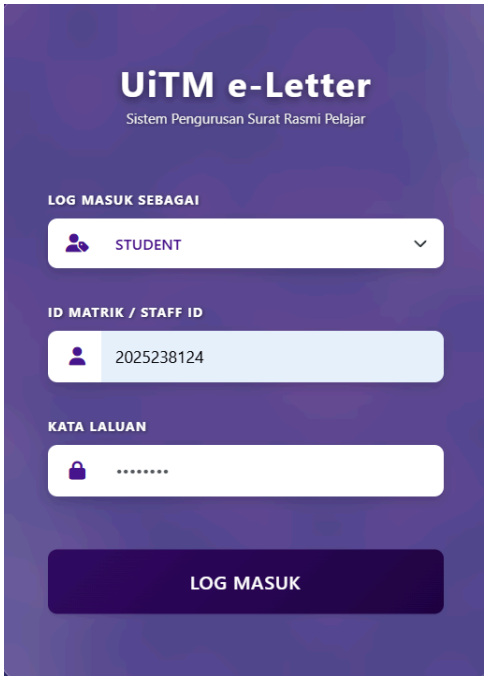


Figure 8: User Authentication Interface

Key Functions

Verify User Login: The function checks user authentication prior to the system use. It ensures that all the functions and features of the system are used by registered users only.

Role-Based Access Control: The system determines if the user is a student or administrator and shows a student and administration dashboard separately.

User Access Management: The function regulates user access to ensure no malicious use of crucial information happens.

Benefits: Protects confidentiality and secures access to student data. This feature provides organization to the system as all student functions are segregated from administrator functions.

6.2 Student Letter Request Submission

Letter Request Submission enables students to submit online the official letter application form to the system. Instead of getting the letter form drafts ready by doing them using the Microsoft Words or some other tools, students could key in details to the system. After the information has been submitted, it will save in the database and be forwarded to the HEP administrators for approval and to process with the letter.

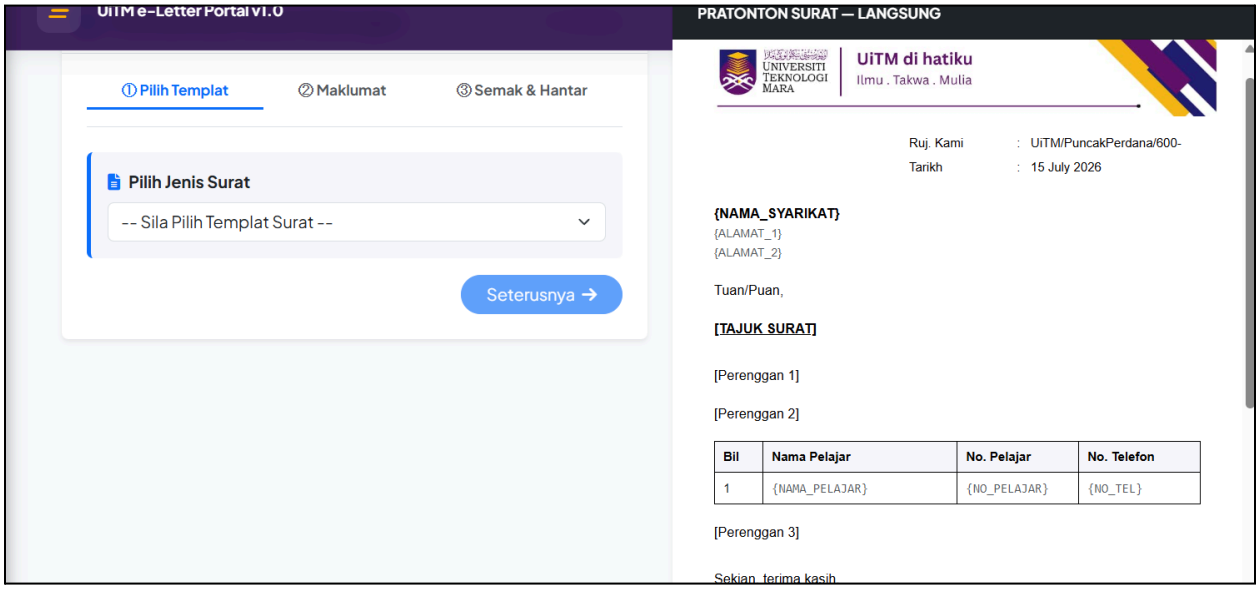


Figure 9: Student Letter Request Submission Interface

Key Functions

Digital Letter Application Form: Enables students to fill the required detail necessary for letter preparation, including student's information, activity, and the intent behind the letter.

Application Submission: Students submit the filled out application on the system directly instead of using any third party system.

Information Storage: The submitted details of the requests are stored in the database.

Benefits: This reduces the amount of paperwork involved, minimizes applications that have incomplete information and makes the entire process of submitting the letter easier and quicker.

6.3 Letter Request Management

The Letter Request Management system provides the HEP administrators to view and control the whole applications which students submit via a central point. The HEP administrators will control the submitted students requests and track the processing stages of student applications. HEP administrators are able to review the student request submitted, then they have to take the actions required before they will have to approve the request.



Figure 10: Letter Request Management Interface

Key Functions

View Submitted Request : administrators are able to see every student request who has applied through the system .

Review Application Information : Administrators are able to view the submitted info in order to know if they've supplied enough information.

Update request status : Administrators are able to change the status of each request which are Pending, Accepted, or Rejected.

Benefits : This feature helps admins become more efficient by diminishing the human processes that need a lot of reviewing .

6.4 Letter Template Management

The feature for Managing Letter Template enables the administrators to control all the official letter templates. Standardized letters are generated with the help of the official letter templates so that the issued letters do not deviate from the set norms. This feature saves administrators from drafting all letters from scratch.

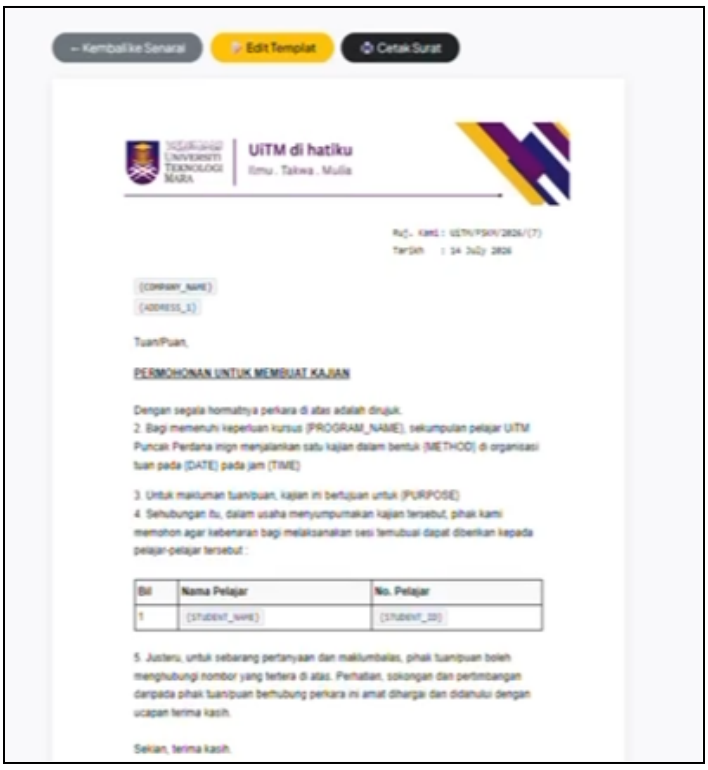


Figure 11: Letter Template Management Page

Key Functions

- **Template Storage :** This will allow the administrator to save the formats of the official letters in the system.
- **Template Selection :**It will enable the administrator to pick the correct format from the saved templates according to the type of letter the users ask for.
- **Template Modifications :** The administrator will be able to modify or change the saved format when needed.
- **Benefits :** By using this feature the time will be saved and all official letters should look identical.

6.5 Automated Letter Generation

Automated Letter Generation allows the system to automatically create a formal letter after the request has been assessed and authorised by the relevant administrators. The letter content is created by fetching student's request data and integrating this data along with the chosen letter template.

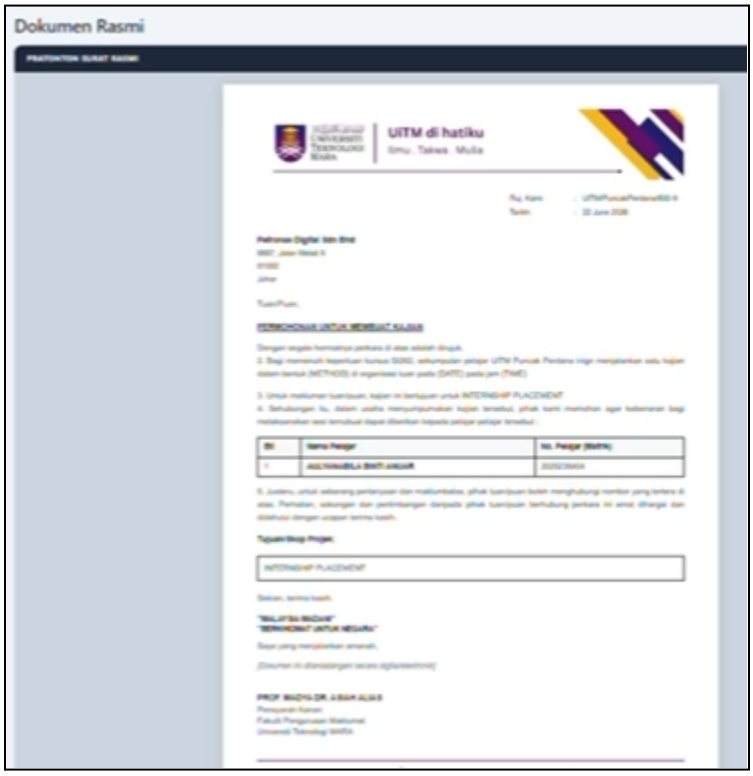


Figure 12: Generated Letter Output

Key Functions

- **Data Retrieval :** the system extracts the approved student data from the database.
- **Document Generation :** The system uses the template which it had previously generated to auto-generate the final official letter.
- **Final Letter Output :** A system generated output of the finalized official letter is prepared for display, saving and distribution to the student.
- **Benefits :** It saves time and labor of office employees which helps minimize mistakes in documents whereas avoidance of manual effort.\

6.6 Application Tracking

With the Application Tracking feature students are able to keep track of what they have already completed as well as keeping up with what needs to be done for their letter requests. Other than that, the students will not have to follow up on their requests by reaching out to HEP staff as they are now able to view their application updates independently. This will enable students to verify their request is either pending review, has been approved, or is finalized.



Figure 13: Application Tracking Page

Key Functions

- **Status Monitoring :** Provides students with access to check the current status of their application.
- **Request History :** Keeps a record of your applications so they can be referenced at any time for future use.
- **Letter Availability Checking :** Notifies or displays when the generated letter is ready.
- **Benefits :** It improves interaction between students and HEP staff, and gives more control to the student in the management of application.

7.0 Workflow of Form

This section explains the implementation of the **CRUD operations**, which consist of Create, Read, Update, and Delete, in the UiTM e-Letter Portal. Both **Administrator** and **Student** users are provided with similar features for managing letter requests. However, administrators can manage all submitted requests, while students can only manage their own requests.

7.1 View Letter Requests

The figure shows the list of submitted letter requests. Users can view important information such as the students name, organisation,programme, status, and submission date.

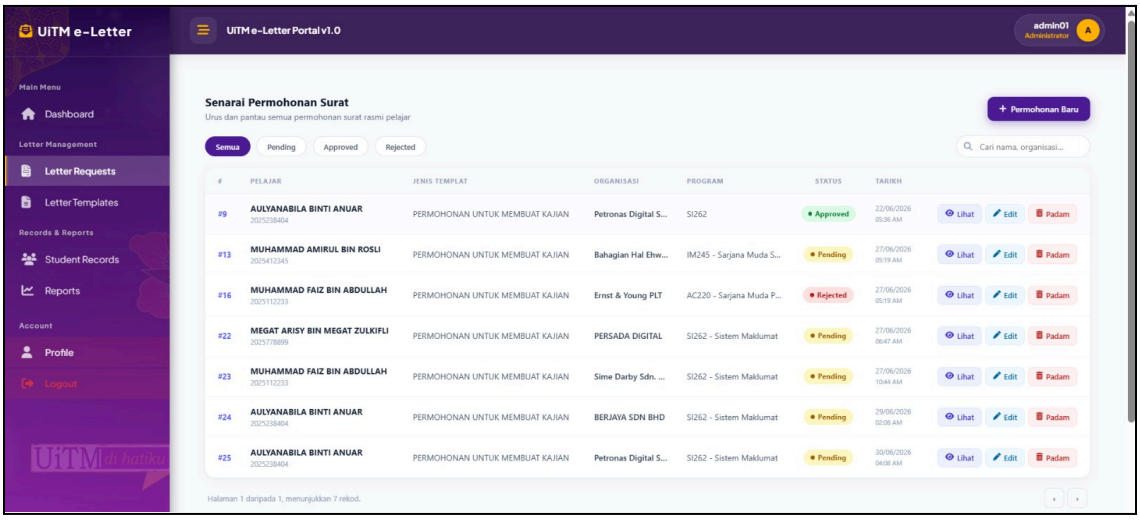


Figure 14: View Letter Requests

The available functions include:

- Filtering requests by **Pending**, **Approved**, or **Rejected**
- Searching for a specific request
- Selecting the **View**, **Edit**, or **Delete** action

Administrators can view all requests, while students can only view their own requests.

CRUD Operation: Read

7.2 Update Letter Request

The figure shows the form used to update a letter request. Users can edit information such as the programme name, organisation, address, and purpose of the request. A live preview is displayed on the right side to show the updated letter format before the changes are saved.

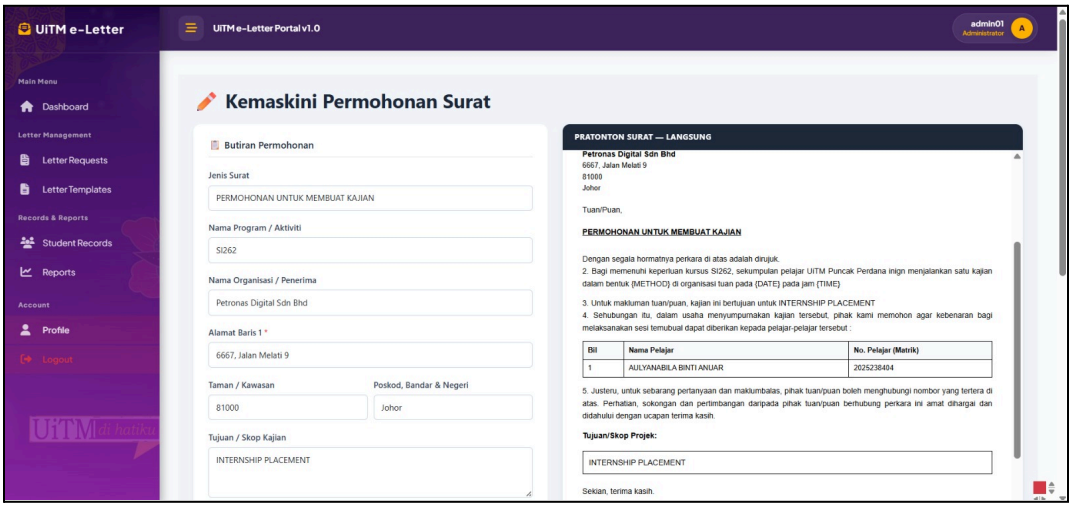


Figure 15: Update Letter Request

This feature is available to both administrators and students according to their access permissions.

CRUD Operation: Update

7.3 Delete Letter Request

The figure shows the confirmation message that appears before a request is deleted. The user must confirm the action before the system permanently removes the selected record.

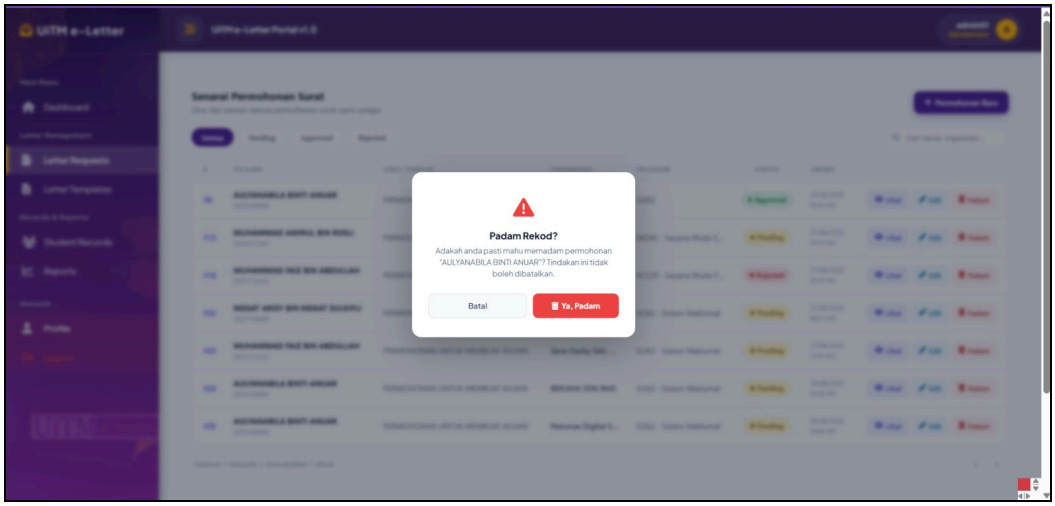


Figure 16: Delete Letter Request

This confirmation process helps to:

- Prevent accidental deletion
- Inform the user that the action cannot be undone

Both administrators and students will receive the same confirmation message when deleting a request.

CRUD Operation: Delete

7.4 View Letter Templates

The figure shows the list of letter templates available in the system. Administrators can search for a template and select the appropriate action to view, edit, or delete it.

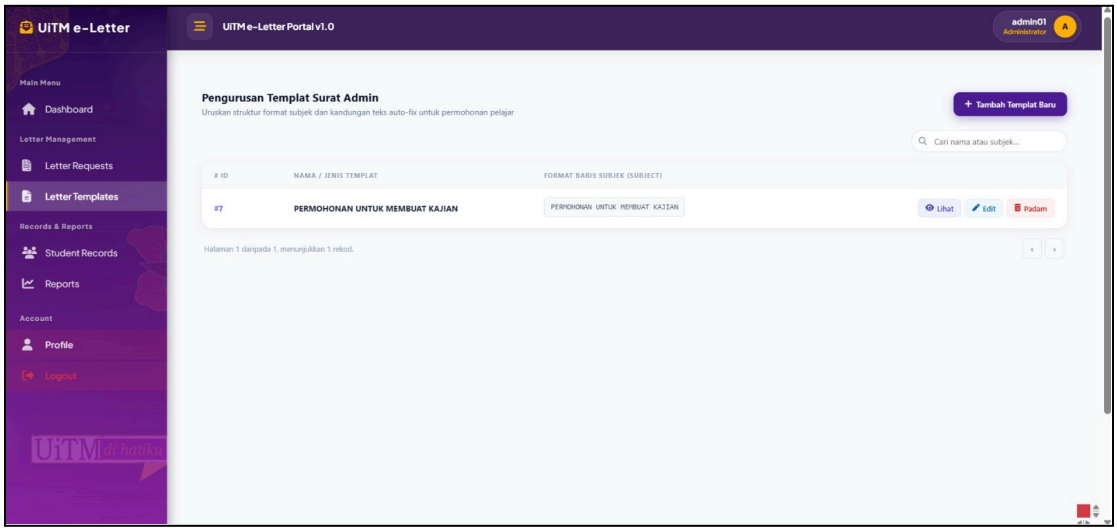


Figure 17: View Letter Templates

This feature is restricted to administrators because letter templates are used as the official format for generating student letters.

CRUD Operation: Read

7.5 Create a New Letter Template

The fifth figure shows the form for creating a new letter template. The administrator is required to enter details such as:

- Template title
- Recipient information
- Subject line
- Letter content

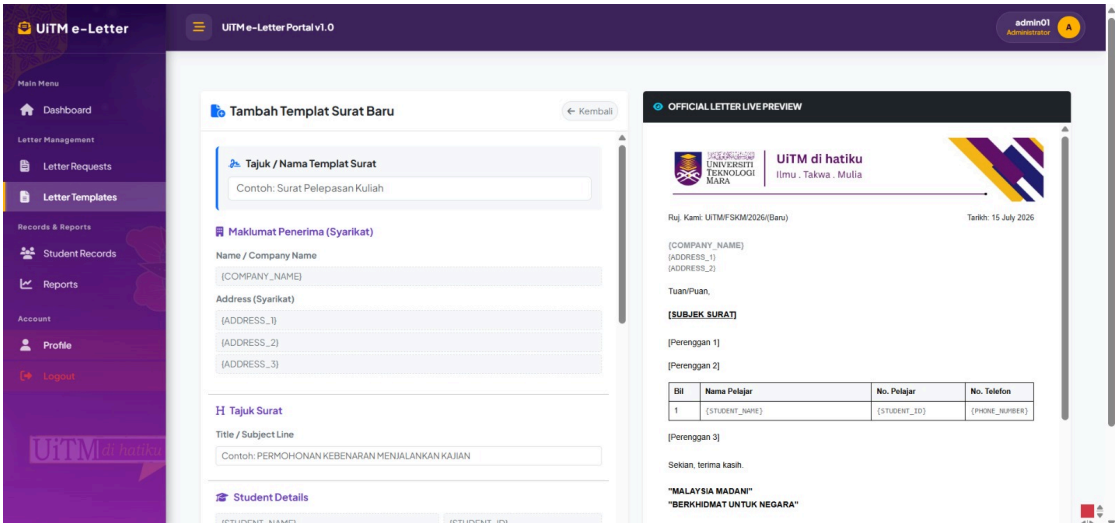


Figure 18: Create a New Letter Templates

A live preview is provided on the right side to help the administrator review the letter layout before saving the template.

CRUD Operation: Create

7.6 Update Letter Template

The figure shows the interface for updating an existing letter template. The administrator can modify the template name, recipient placeholders, subject, and paragraph contents. Any changes made will be reflected automatically in the live preview.

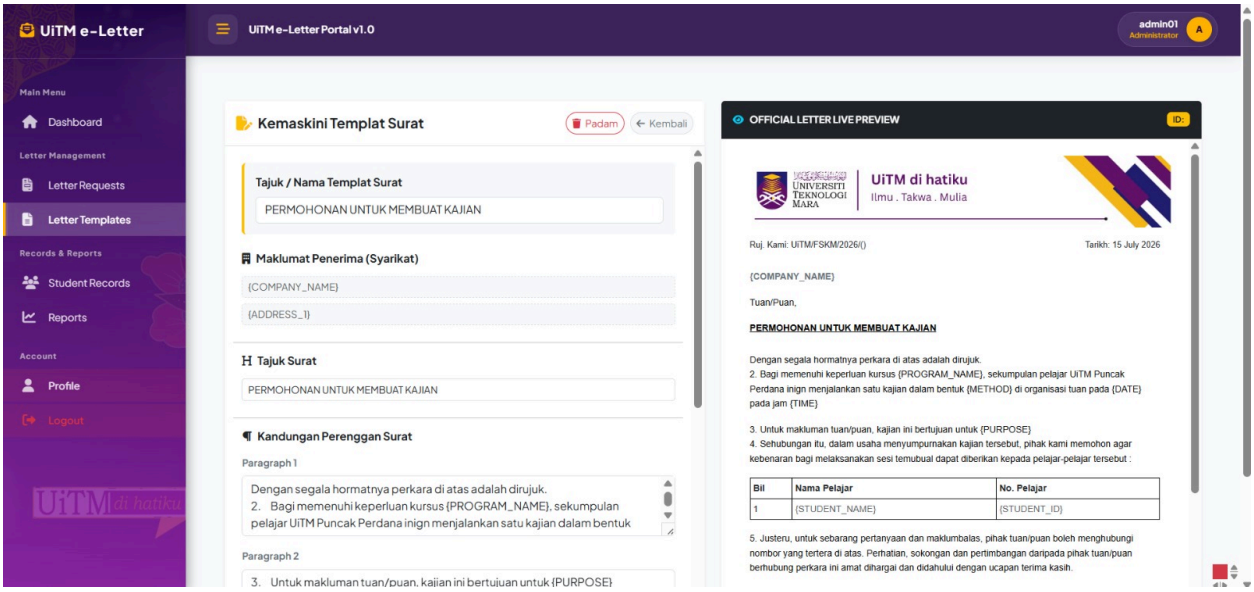


Figure 19: Update Letter Template

This allows the administrator to verify the updated format before saving it into the system.

CRUD Operation: Update

7.7 Delete Letter Template

The figure shows the delete confirmation dialog for a letter template. The administrator must confirm the action before the template is permanently removed.

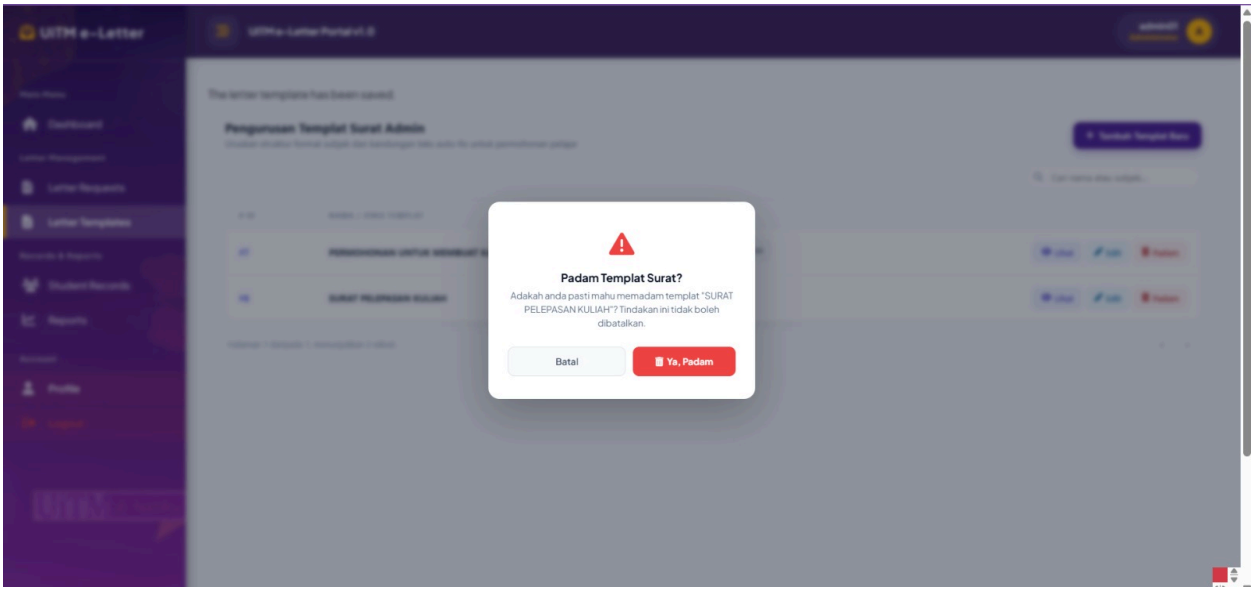


Figure 20: Delete Letter Template

The confirmation message is included to prevent important templates from being deleted unintentionally.

CRUD Operation: Delete

8.0 Team Roles & Contributions

The development of the website was divided into two main sections, the Admin Section and the User Section. Each team member was assigned specific responsibilities to ensure that the development process was well organised, efficient, and completed according to the project requirements.

Section	Team Member	Assigned Modules/ Responsibilities	Main contribution
Admin	Aulyanabila binti Anuar	Development of the Admin section and support for the integration of administrative functions	Assisted in developing, coordinating, testing, and ensuring the smooth operation of the Admin section
Admin	Nurhanisah Basyirah binti Kamaruddin	Letter Templates, Student Records, Generated Letters, and Admin Profile	Designed and developed the assigned Admin modules, including their interfaces, functions, and data management features
User	Rabiatul Adawiyah binti Samsuri	Development of the User section and support for the integration of user-related functions	Assisted in developing, coordinating, testing, and ensuring the usability of the User section
User	Raja Ellya Atassya binti Raja Fairulnizam	My Request, User Profile, and Generated Letters	Designed and developed the assigned User modules to allow users to manage requests, update their profiles, and access generated letters

Overall Team Contribution

All team members contributed to the planning, development, testing, debugging, and improvement of the website. Regular discussions and collaboration were carried out to ensure consistency between the Admin and User sections.

9.0 Contact Information

The following table provides the contact details of the project support team. Users may contact any of the team members for assistance regarding website access, system functionality, technical issues, or general enquiries.

No	Support Team Member	Contact Number	Email Address
1.	Nurhanisah Basyirah binti Kamaruddin	011-25407759	Kamaruddinhanisah@gmail.com
2.	Aulyanabila binti Anuar	011-28684469	AulyaNabila@gmail.com
3.	Rabiatul Adawiyah binti Samsuri	011-33915514	AdawiyahRabiatul@gmail.com
4.	Raja Ellya Atassya Binti Raja Fairulnizam	010-24443030	Ellyaattasya@gmial.com

For support or further enquiries, users are encouraged to provide a clear description of the issue when contacting the support team. This will help the team identify the problem and provide an appropriate response more efficiently.

10.0 Conclusion & Reflection

The UiTM e-Letter Portal was developed to make the process of requesting and managing official university letters easier and more organised. Students can submit requests, check their application status, update information, and access generated letters. Administrators can review requests, manage records, and maintain letter templates.

The system also provides useful features such as search, filtering, live previews, dashboard summaries, and confirmation messages. These features help users complete their tasks more efficiently and reduce errors.

Throughout the project, the team gained practical experience in system development, including planning, interface design, database connection, feature development, and testing. The division of tasks between the Admin and Student sections helped each member focus on their responsibilities while working together to integrate all modules.

The project also improved the team's communication, teamwork, problem-solving, and time-management skills. Overall, the UiTM e-Letter Portal achieved its objective of providing a simple and systematic platform for managing official letters. The experience gained will be useful for future projects and professional work in information technology.